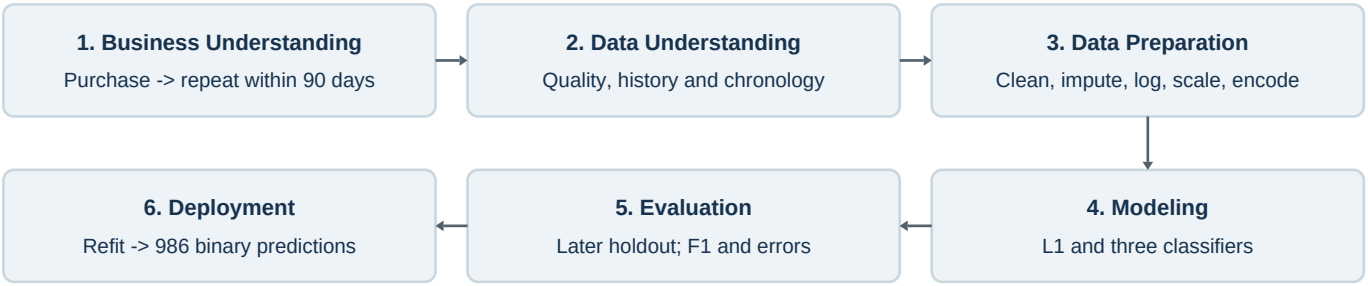
Predicting Repeat Purchases

Machine Learning 2026/27 | Homework | Group 06 | 19 October 2026



1. Goal and data

Predict whether a customer buys again within 90 days of the current purchase. Each row is a purchase occasion; customers can repeat. **Binary F1** is the official Kaggle metric. The supplied files have 6,534 training purchases, 986 test purchases and 1,143 customers. A left join through customer_id covers every purchase.

2. Preparation and rationale

Remove 10 exact duplicates excluding ID and 55 distinct undated records with no purchase information, leaving **6,469 dated training purchases**. Preserve all test IDs and the original files. Exclude purchase/customer IDs, empty customer_type, company name, address and selected granular geography. Geographic codes are categories, not distances.

Invalid negative history values and suspected exact 1e9 placeholders become missing values. Reconstruct historical averages as total/count, assuming these totals and counts are reliable; zero counts leave the mean undefined. Mark impossible historical maxima and annual spend as missing. Retain coherent high amounts.

Fill remaining numeric gaps with training medians and keep missing-value flags; text gaps become Missing. Apply **log1p to monetary inputs, then standard scaling**. One-hot encoding pools categories with fewer than 20 training observations. Include candidate year/month/weekday inputs while recognising that dates are anonymised.

3. Temporal assessment

Reserve **1,298 later purchases from 17 June 2023** as the outer holdout. Development has 5,024 purchases ending 18 March 2023; the intervening 147 purchases form a 90-day label-maturity gap. Same-day purchases stay together. Repeated customers are allowed because the task includes later purchases by known customers.

Inside development, use three expanding temporal folds, each with its own 90-day gap. Fit imputation, scale, category frequencies and supervised selection **only on each training fold**. Select the classifier and feature strategy by mean inner F1 before assessing the later holdout. Use the default decision rule; no threshold is chosen after viewing holdout results.

The six notebooks follow CRISP-DM and are executable in order. Learned preparation is repeated through three small shared functions; the individual operations remain visible in Data Preparation.

4. Selection and classifier comparison

Compare all inputs with L1-logistic selection (C=0.1; coefficient tolerance 1e-6), fitted within each fold. Candidates: logistic regression (C=1), depth-5 tree (minimum leaf 20), and 200-tree random forest (minimum leaf 5). Seeds are fixed. This compact homework does not perform a broad parameter search.

Candidate / feature strategy	Mean inner F1	Fold SD
Baseline / All	0.7044	0.0243
Logistic regression / L1	0.7321	0.0209
Decision tree / All	0.7181	0.0209
Random forest / L1	0.7384	0.0096

Forest + L1 wins the predefined mean-F1 rule. Forest without selection scores 0.7368: the 0.0016 difference is small and not conclusive. L1 reduces full-development inputs from **276 to 54**. random_noise remains under the fixed rule; this does not prove stable signal. Targeted ablation is left for the final project. Fold SD is descriptive, not a confidence interval.

5. Fixed later holdout

Solution	F1	Precision	Recall	Accuracy
Baseline	0.7525	0.6032	1.0000	0.6032
Forest + L1	0.7778	0.7869	0.7688	0.7350

The F1 gain is **2.53 percentage points**. Confusion counts: TN=352, FP=163, FN=181, TP=602. The baseline predicts repeat purchase for everyone; the forest improves precision while retaining recall of 0.769. Costs and margins are not supplied, so no profit estimate is made.

F1 is 0.7910 for customers seen in development versus 0.6230 for those unseen. Of the latter group's 145 purchases, 101 already have history. Separately, 44 first purchases contain 12 positives, none identified (F1=0): a cold-start weakness with a small sample. No model changes follow holdout inspection.

6. Refit, export and limits

Refit the same method on all **6,469 eligible labelled purchases**, including the former gap. Test starts 98 days after the last training purchase, so all training targets are mature. Training-only preparation yields 302 inputs; L1 retains 58 before fitting the same forest settings.

ML_Group_06_Version01.csv has exactly 986 original IDs and binary predictions: 645 positive, 341 negative. It contains only ID and repeat_purchase_90d, without an extra index. These counts are not test performance. The six notebooks reproduce the file. Kaggle upload and its public score remain pending.

The earlier group EDA used the full training file, so this holdout is not completely untouched. There is one outer period; customers repeat across rows; cleaning assumes reliable totals/counts; customer attributes lack historical snapshots. Calendar and rare-category choices need further study. The final project will add five taught algorithms, optimisation and an additional investigation.

Sources: course homework/project briefs and feature-selection/model-assessment materials; supplied CSVs; Kaggle Overview, Data and Rules screenshots provided on 19 October 2026. No external research or public leaderboard score is used.